Energy Load Forecasting Proposal:
A Data-Driven Approach to Demand Forecasting in a Deregulated Energy Market

Analytics Team 5 (HW 5)

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Overview

American Electric Power Co. (AEP) is seeking analytical proposals to forecast total daily energy loads for its Appalachian Power Territory (AEPAPT) to stay competitive and operate efficiently in an increasingly deregulated energy market. Analytics Team 5 has developed and tested both an Exponential Smoothing model (ESM) and a seasonal ARIMA (SARIMA) model to predict daily energy demand for the AEPAPT region, comparing their performance using standard accuracy metrics. Our analysis shows that the SARIMA model delivers the most accurate and reliable forecasts, with predictions that are within approximately 7% of actual daily demands, on average. This level of accuracy provides AEP with a strong foundation for more effective planning, budgeting, and peak-demand management.

Methodology & Analysis

This section describes the data preparation and modeling process, including the datasets used, the selection of the best ESM and SARIMA models, and the performance results of both models.

Data Used

The dataset, sourced from PJM's Data Miner, provided the hourly energy load in megawatts (MW) for the AEPAPT Region, which we rolled up to total daily energy load in MW. The training dataset comprised 1,857 observations from August 1, 2018, to August 31, 2023, and the validation dataset consisted of 366 observations from September 1, 2023, to August 31, 2024.

ESM Model Selection

To identify the most suitable ESM model for forecasting total daily energy load, we fitted several standard candidate models to the training data, including simple exponential smoothing, Holt's linear trend, Holt's damped trend, and both additive and multiplicative Holt-Winters models. Each model can be represented in terms of its error, trend, and seasonality components, as reported in Table 3 of the appendix. We also employed an automated search procedure (ETS), which systematically evaluated all component combinations. We assessed each model's fit using the corrected Akaike Information Criterion (AICc), with the results presented in Table 1 in ascending order.

Table 1: AICc Comparison of ESM Models

Model	AICc
ETS (Search)	46886.55
Additive Holt-Winters	46930.92
Multiplicative Holt-Winters	46948.04
Simple Exponential Smoothing	47299.83
Holt's Damped Trend	47306.13
Holt's Linear Trend	47313.64

As shown in Table 1, the ETS search produced the best-performing model with the lowest AICc value of 46886.55. The ETS search model was characterized by a multiplicative error, no trend, and additive seasonality, and was selected for final forecasting.

SARIMA Model Selection

We built a SARIMA model in addition to the ESM model to compare forecasts of total daily energy load. We started by addressing both weekly and yearly seasonality using Fourier terms. We selected the number of weekly Fourier terms using a grid search and yearly Fourier terms using a plot of the relationship between the number of terms and goodness of fit, as reported in Figure 3 of the appendix. We selected K = 3 for the weekly Fourier and K = 2 for the yearly Fourier, which the plot identified as an effective balance between model fit and complexity. After fitting these terms, we added one non-seasonal difference based on the results of a Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test on residuals.

We examined the correlation plots, autocorrelation function (ACF) and partial autocorrelation function (PACF), of the differenced Fourier model to determine which autoregressive (AR) and moving average (MA) terms would be suitable for testing in the model. The PACF spikes suggested a lag of close to 4 for non-seasonal AR, and potentially seasonal MA terms as well, as shown in Figure 4 of the appendix. The ACF spikes also suggested a small lag of non-seasonal MA, and that seasonal MA lags may be appropriate, as shown in Figure 5 of the appendix. We tested all combinations of candidate models with non-seasonal AR3-4 and MA1-2, as well as seasonal AR1-2 and MA1-2. These candidate models were compared using AICc to evaluate their goodness of fit, and the top five models were then further examined for accuracy on validation data to ensure generalizability to a hold-out dataset. We then ran a Ljung-Box test for white noise on the top-performing candidate model, which confirmed the model successfully captured autocorrelation in the data. This selection process identified the SARIMAX(3,1,1)(2,0,2) model, with weekly (K = 3) and yearly (K = 2) Fourier terms, as the best SARIMA model for forecasting total daily energy load, based on its strong goodness of fit and accuracy.

Results

This section summarizes and compares the performance of the ESM and SARIMA models on the held-out validation dataset using both accuracy metrics and visual assessments.

Model Performance – Accuracy Metrics

We assessed the forecasting performance of the best ESM and SARIMA models using the validation dataset. To capture both absolute and relative forecast accuracy, we reported the mean absolute error (MAE) and the mean absolute percentage error (MAPE). Table 2 summarizes these results, where smaller MAE and MAPE values indicate higher forecasting accuracy.

Table 2: Forecasting Accuracy of the ESM and SARIMA Models

Model	MAE	MAPE
ESM	11,332.69	10.23%
SARIMA	7,296.90	6.81%

The SARIMA model outperformed the ESM model on both accuracy metrics presented in Table 2. The SARIMA model achieved an MAE of 7,296.90, indicating that its daily forecasts deviated from actual total energy loads by approximately 7,300 MW on average. A MAPE of 6.81% suggests that the SARIMA model's predictions were, on average, within roughly 7% of the actual daily load values. In comparison, the ESM had larger errors, with an MAE of 11,332.69 and an MAPE of 10.23%. These results indicate that the SARIMA model provides more accurate and reliable forecasts of total daily energy load.

Model Performance – Visual Assessments

We can further assess the forecasting performance of the best ESM and SARIMA models by comparing their forecasted total daily energy loads to the actual observed values on the held-out validation dataset. Figure 1 illustrates this comparison for the ESM model, showing how well the model's forecasts align with the actual total daily energy load patterns between September 1, 2023, and August 31, 2024.

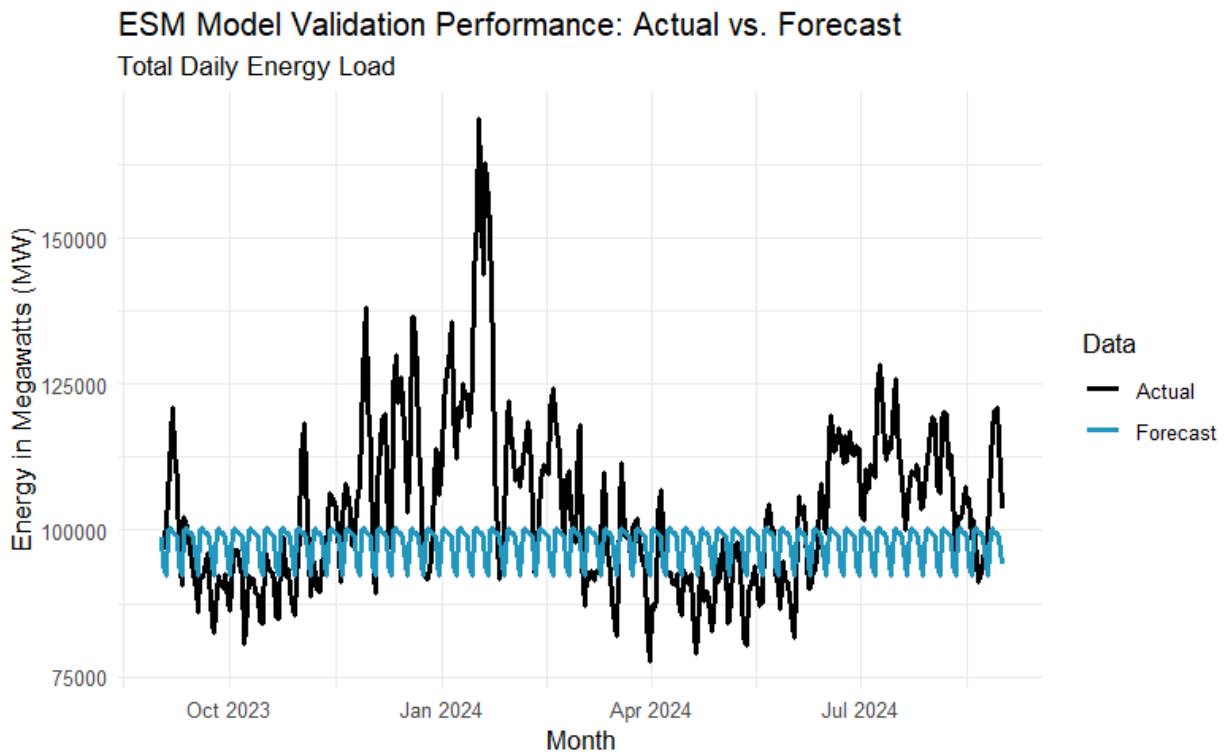

Figure 1: ESM Model Validation Performance: Actual vs. Forecast

Figure 1 displays the ESM model's forecast performance by comparing actual daily energy load (shown in black) with the model's predicted values (shown in blue) during the validation period. The x-axis represents the time, and the y-axis shows the total daily energy load in MW. The ESM model captures the regular weekly fluctuations in demand but fails to reflect the larger swings associated with yearly seasonal patterns, particularly the winter peak. This limitation is expected, as traditional ESM models are not designed to handle multiple or complex seasonalities. Nevertheless, the ESM model still provides a valuable baseline against which other approaches, such as SARIMA, that can model more complex seasonal patterns can be evaluated.

Likewise, Figure 2 shows the SARIMA model's forecasts compared to the actual total daily energy load on the held-out validation dataset.

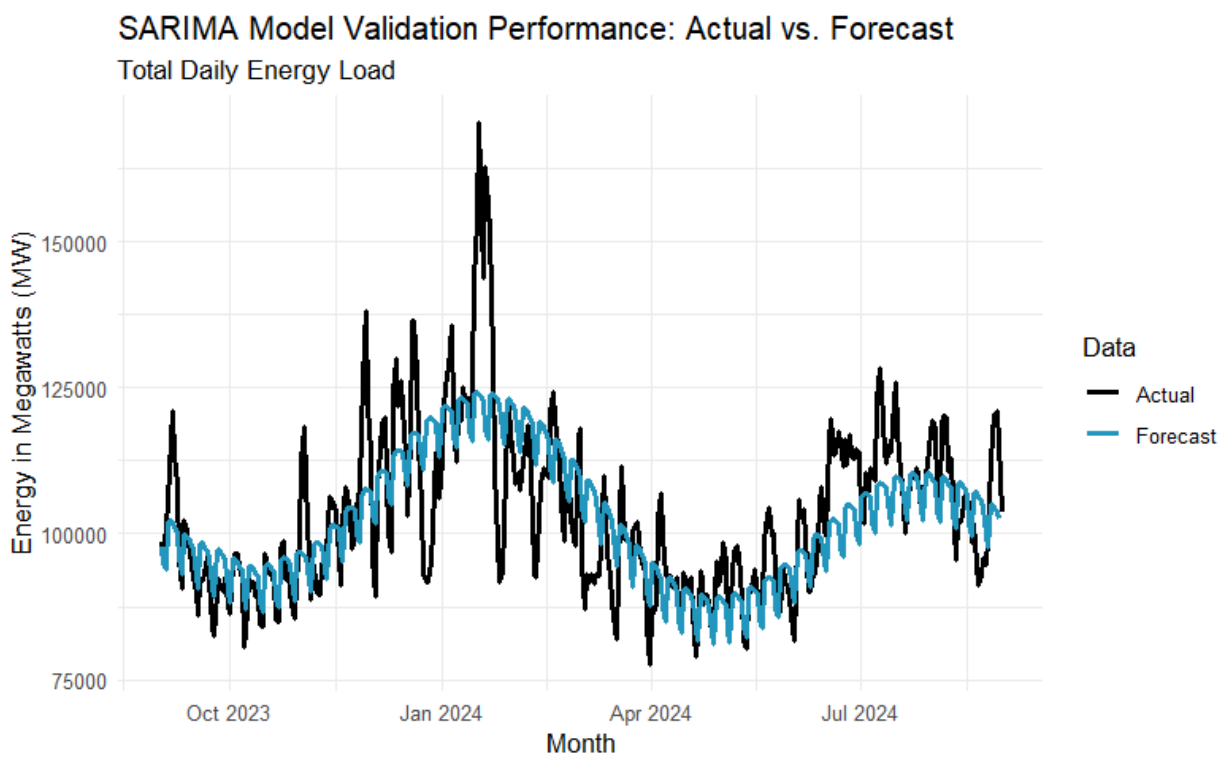

Figure 2: SARIMA Model Validation Performance: Actual vs. Forecast

Figure 2 presents the SARIMA model's forecast performance by comparing the actual daily energy load (shown in black) with the model's predicted values (shown in blue) during the same validation period. As in the previous figure, the x-axis represents time and the y-axis shows total daily energy load in MW. Unlike the ESM model, the SARIMA model provides a much closer alignment between the actual and forecasted total daily energy loads. Furthermore, the SARIMA forecasts successfully capture both the weekly fluctuations and the broader yearly seasonal pattern, including the more pronounced winter peak. This improved alignment demonstrates the SARIMA model's ability to model multiple and more complex seasonalities, resulting in forecasts that more accurately reflect the underlying behavior of energy demand in the AEPAPT region.

Recommendations

Based on its strong forecast accuracy and ability to capture complex seasonal patterns, we recommend that AEP adopt the SARIMA model to forecast total daily energy demand in the AEPAPT region. To further improve forecast reliability in an increasingly dynamic and deregulated energy market, we suggest the following:

1. Incorporate external factors, such as weather and economic indicators, to better capture day-to-day variations in demand.
2. Update the model regularly to ensure forecasts reflect evolving consumption patterns and regulatory changes.
3. Leverage forecasts to support strategic decision-making, including budgeting, peak-demand procurement, resource scheduling, and market participation, particularly as competitive pressures increase. Based on our forecasts, this includes preparing for higher operational expenses during the peak winter and summer months.

4. Consider exploring real-time or intra-day forecasting enhancements, resources permitting, which may become more critical if short-term market responsiveness becomes increasingly significant under accelerated energy market deregulation.

This forecast will enable AEP to anticipate seasonal variations in energy demand and maintain high customer satisfaction, a growing priority as more states move toward energy deregulation.

Conclusion

Reliable energy-load forecasting is essential for efficient operations in the AEPAPT region, particularly as the market continues to deregulate. The SARIMA model demonstrated the highest accuracy and the strongest ability to capture both weekly and yearly demand patterns, with forecasts that are within approximately 7% of actual daily energy loads, providing a robust foundation for planning and operational decision-making. By adopting our SARIMA model, AEP can better anticipate customer demands, manage costs, and maintain reliable service across its network. Going forward, regularly updating and retraining the model will ensure it continues to reflect evolving energy usage patterns and system conditions.

Appendix

Table 3: Error, Trend, and Seasonality Components of Candidate ESM Models

Model Name	Error	Trend	Seasonality
Simple Exponential Smoothing	Additive	None	None
Holt's Linear Trend	Additive	Additive	None
Holt's Damped Trend	Additive	Additive (Damped)	None
Additive Holt-Winters	Additive	Additive	Additive
Multiplicative Holt-Winters	Multiplicative	Additive	Multiplicative
ETS (Search)	Multiplicative	None	Additive

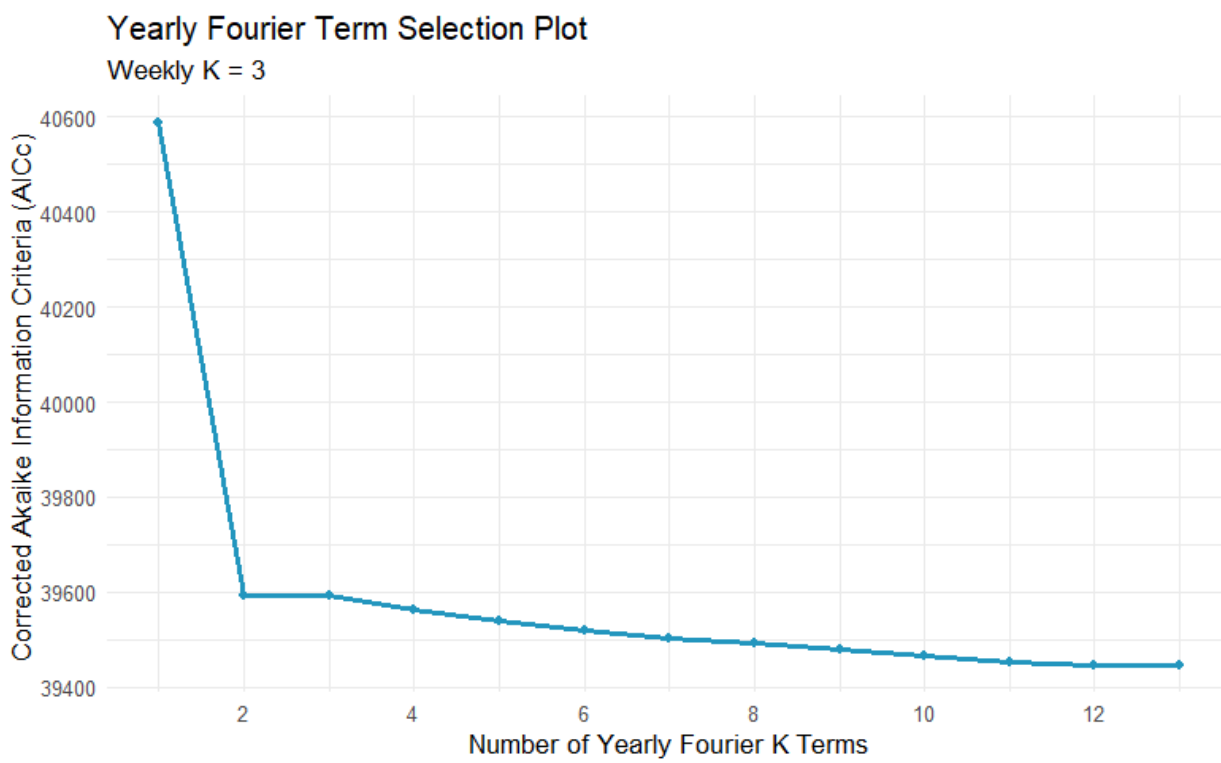

Figure 3: Yearly Fourier Term Selection Plot

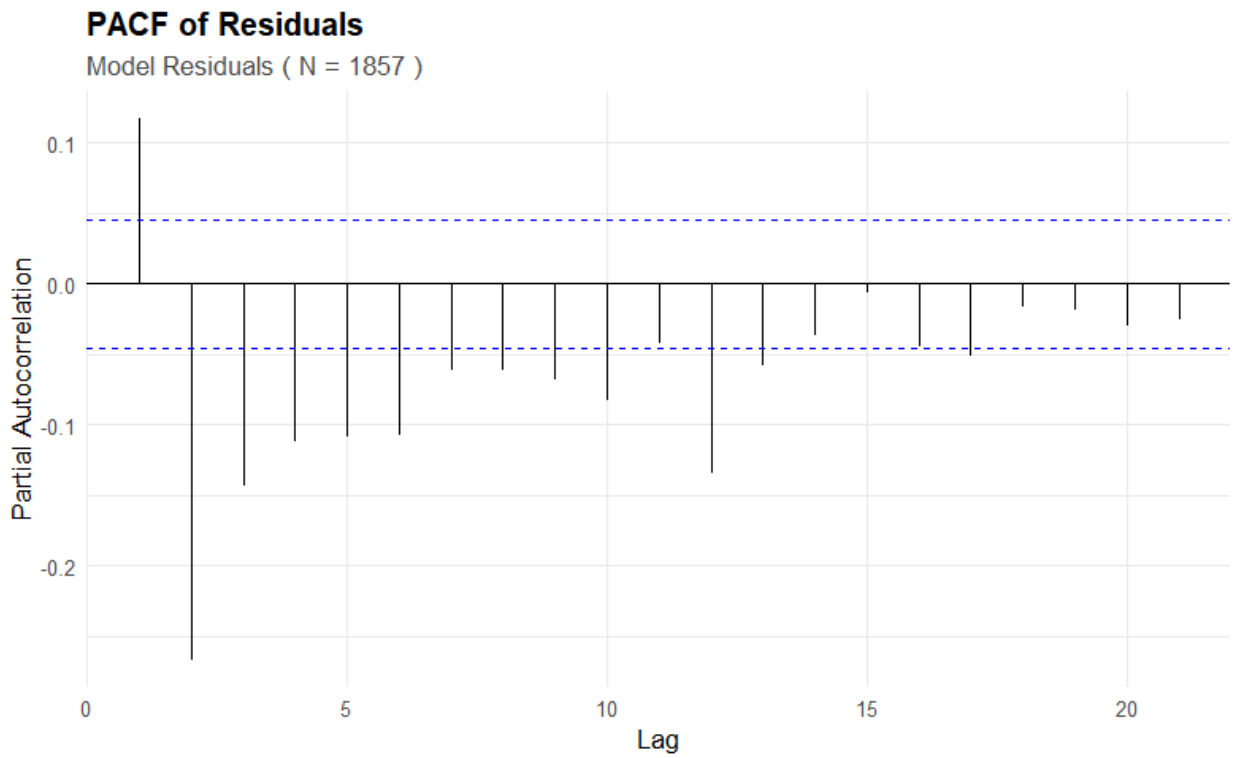

Figure 4: PACF of Residuals

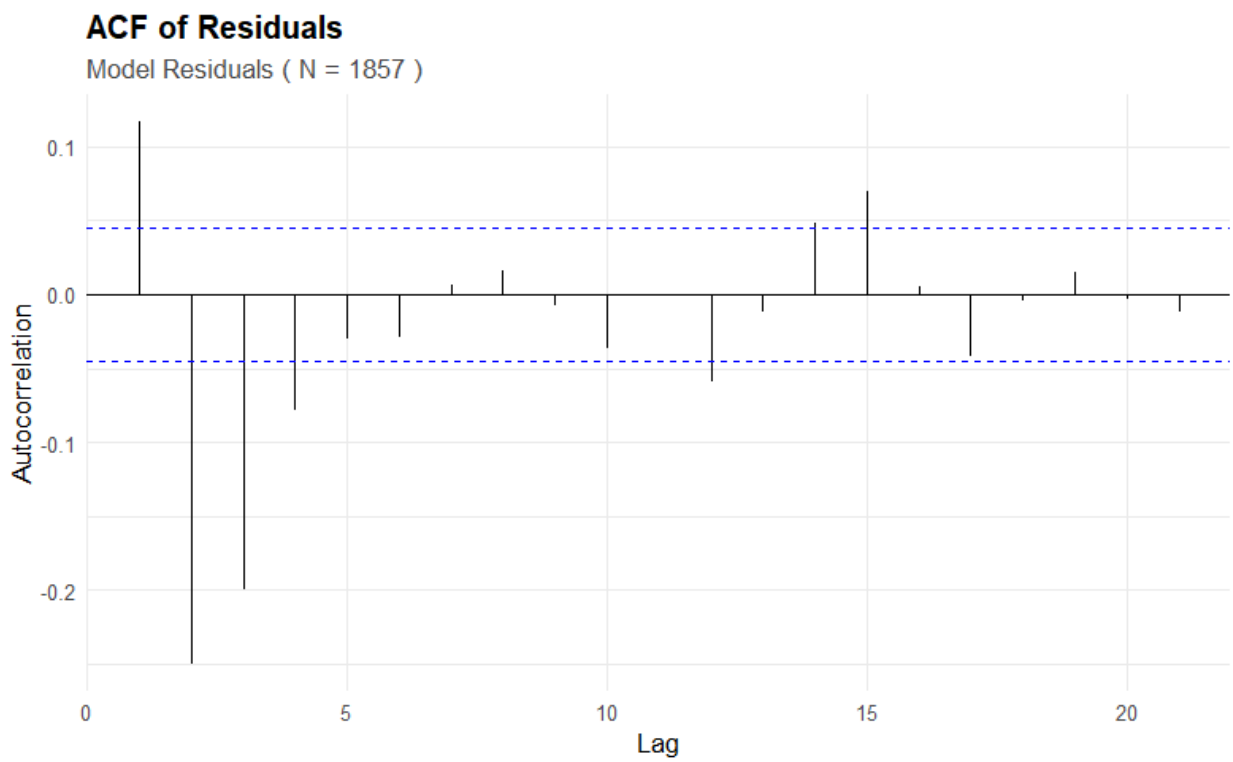

Figure 5: ACF of Residuals

Homework Report Checklist

As instructed by Hiya and Dr. Egan Warren, the team member(s) responsible for checking each item should enter their initials in the field next to each question. All items should be addressed before submitting the assignment with the initialed checklist attached.

Sections & Structure

Overview

SAM	Is the overview concise?
SAM	Does it provide context about the business problem? <Content>
SAM	Does it briefly address your team's work, quantifiable results, and recommendations? <Action>
SAM	Does it offer audience-centered reasons for recommendations? <Context>

Body Sections

TL	Does the report body include information on methods, analysis, quantifiable results, and recommendations?
TL	Is content grouped into appropriate sections (*methodology*, *analysis*, *results*, *recommendations*)?

Conclusion

AG	Does the report have a conclusion?
AG	Does the conclusion sum up the report and emphasize relevant takeaways?

Structure

TKP	Does each major section have a heading?
TKP	Are sections, subsections, and paragraphs organized logically for easy navigation?

Visuals

Introduction, Discussion, and Captions

BMH	Is each visual introduced in the text before it appears?
BMH	Is each visual close to where it is introduced?
BMH	Does each visual include a title with the following information: type (*table* or *figure*), number, and a descriptive caption?
BMH	Is each visual discussed and interpreted in the text?
BMH	Are figures and tables numbered separately?
BMH	Are table captions above the table? Are figure captions below the figure?

Visual Design

SAM	Do figures/tables use audience-friendly labels rather than variable names?
SAM	Are the visuals easy to interpret?
SAM	Are the visuals appropriately sized?
SAM	Do tables appear on one page (*not split between 2 pages*)?
SAM	Are legends and axis labels included for figures?
SAM	Are numbers in tables right aligned?
SAM	Are the visuals designed well (*ex*: *re-created in Word or Excel*, *not blurry or stretched*,…)?

Document Design

Title Page Design

TKP	Does it include a descriptive title?
TKP	Does it state the team name, team members' names, and the submission date?

Table of Contents Design

TL	Does it list all the major sections of the report with corresponding page numbers?
TL	Do the page numbers and sections in the Table of Contents match the report?

Document Design for Entire Report

AG	Is a standard typeface (*Calibri*, *Arial*, *etc.*) used?
AG	Is the size of the body text between 10-12 pt.?
AG	Are headings and subheadings used to organize information?
AG	Are distinctive text styles (*bold*, *italic*, *etc.*) used to distinguish between heading levels?
AG	Are text styles for headings used consistently (*ex*: *all level-one headings are bold*)?
AG	Are all paragraphs an appropriate length (*fewer than 12 lines*)?
AG	Is white space used to indicate paragraph breaks?
AG	Are bullet lists used for a series of items and numbered lists to show a hierarchy?

Writing Style and Mechanics

Spelling and Capitalization

BMH	Are spelling errors located and corrected?
BMH	Is spelling consistent throughout (*no switching between acceptable spellings*)?
BMH	Is capitalization used appropriately (*proper nouns*, *etc.*)?
BMH	Is capitalization of words consistent throughout the report?

Grammar and Punctuation

SAM	Are verb tenses used appropriately?
SAM	Are marks of punctuation used appropriately?
SAM	Is subject-verb agreement used in every sentence?
SAM	Is the grammar checker updated and are underlined grammar issues addressed?

Writing Style

TL	Are all sentences in the report easy for your audience to understand quickly?
TL	Are most sentences written in active voice?
TL	Are idioms and vague words eliminated from the report?
TL	Are acronyms introduced before being used?
TL	Are well-written topic sentences included at the beginning of each paragraph?
TL	Are lists parallel?
TL	Is the appropriate point of view used when addressing your audience or describing team actions?